

Beyond the Enforcer: Exploring Metaphors for Alternative AI Moderation in Online Games

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Abstract

Online games increasingly rely on AI moderation systems that employ punitive approaches despite player dissatisfaction with these methods. This paper investigates how metaphorical frameworks can inform alternative approaches to AI moderation in gaming spaces. Through four co-design workshops with 16 online game players and AI experts, we explored how stakeholders conceptualise AI moderation beyond traditional punitive metaphors like ‘police officer’ or ‘enforcer’. We present three spotlight designs from these workshops based on the metaphors of the ‘all-seeing eye’, ‘non-player character/NPC’, and ‘cleaner’. Participants’ designs highlighted numerous unexplored applications of AI in the moderation space, including interface and game integration, educational opportunities, and assistive potential. At the same time, these alternative moderation systems raised new challenges and ethical concerns. Drawing from our findings, we present three future design and research avenues for alternative AI-driven moderation in online games, reflecting on potential opportunities and challenges.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

Keywords

Artificial intelligence, moderation, multiplayer games, metaphors, co-design

ACM Reference Format:

Ren Galwey, Lucy A. Sparrow, and Mahli-Ann Butt. 2025. Beyond the Enforcer: Exploring Metaphors for Alternative AI Moderation in Online Games. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720148>

1 Introduction

Online multiplayer games have become virtual communities for millions of players, making effective content moderation crucial

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04
<https://doi.org/10.1145/3706599.3720148>

for ensuring player safety and positive player experiences. While artificial intelligence (AI) has emerged as a crucial tool to assist in the scaling of automated moderation efforts [8], current approaches predominantly rely on punitive measures such as bans, mutes and restrictions [13, 31]. However, this default approach has been shown to frustrate and confuse players, in part due to a lack of transparency, accountability, and opportunities to challenge automated moderation decisions [18, 25]. While operationally beneficial and straightforward for automation, a focus on this approach fails to address underlying issues with online toxicity (the umbrella term given to problematic and disruptive communication and attitudes among players in games), support targets of toxicity or promote community development [15, 20].

The growing sophistication of AI systems, coupled with the evolution of games into vital social spaces [9], presents an opportunity to reimagine online moderation approaches. While research has established the need for more nuanced community governance in gaming spaces [28], little attention has been paid to how metaphorical reconceptualisation might generate alternative AI moderation practices. Drawing from player experience and AI expertise, this paper addresses this gap by co-designing alternative approaches to AI moderation in games using metaphors. Our research questions sought to understand these alternatives and their implications: (1) How do stakeholders (players and AI experts) conceptualise alternative AI moderation in multiplayer games through metaphors?; and (2) What are the potential opportunities and challenges of employing alternative AI moderation systems in online games?

Through analysis of four co-design workshops, we contribute descriptions of participant designs that reflect novel approaches to AI moderation systems alongside key functions, perceptions and challenges. Based on these findings, we propose future design and research directions for alternative AI-driven moderation systems in online multiplayer games.

2 Related work

2.1 AI Moderation in Online Multiplayer Games

Content moderation in online gaming environments has increasingly evolved into a hybrid approach combining human judgment with AI-driven automation [21]. While human moderators are seen to possess comparatively nuanced contextual understandings [25], AI systems offer scalability and the efficient, real-time processing of mass reporting [8]. This has led several large gaming companies to adopt third-party AI-driven moderation solutions such as ToxMod, which provides real-time voice moderation in major games such as *Call of Duty* [3]. Currently, much research in this space focuses



Figure 1: Screenshot from the workshop video demonstrating the fictional game ‘Sorcerers Showdown’.

on increasing detection accuracy, perpetuating a model centred on identifying and disciplining perceived offenders [32].

Studies suggest several limitations with existing AI moderation systems [32]. Players report frustration with opaque decision-making processes and insufficient communication about reports [18, 25]. Furthermore, the punishments associated with these systems fail to create “well-behaved community members”, as players frequently circumvent restrictions or develop new methods of misconduct [16, 18, 25]. The resulting design tension between transparency and misuse exemplifies just one of the many tradeoffs that are encountered in content moderation design [12].

A small body of work has begun to explore the potential value in exploring alternative approaches to moderation. Researchers have investigated frameworks emphasising restorative justice and education over punishment [31] and a support-driven approach focused on helping targets of toxic behaviour [20]. These approaches have further informed AI-driven solutions that are designed to facilitate harassment reporting [11, 27] and foster community [4]. However, the full potential of AI in reimagining online game moderation remains largely underexplored.

2.2 Metaphors in Design

Metaphors are forms of symbolic representations that serve as powerful cognitive bridges that make complex concepts accessible [19]. They are recognised in HCI and design thinking as tools for inspiring creativity, challenging assumptions, better understanding of human relationships to technologies, and helping to bridge the gap between users’ mental models and designers’ conceptual

models [1, 6]. Existing research has explored the role of metaphors in various fields, including moderation [24, 30], robot interaction [1], voice user interfaces [6], and conversational AI agents [14]. Desai and Twidale [6], for example, analysed the metaphors their participants used when referring to voice user interfaces and categorised them according to metaphor type (human, non-human, or fictional), exploited knowledge, origin, and motivation. Their analysis demonstrated how users’ conceptual frameworks influence their technological interactions and expectations.

The domain of content moderation presents a particularly rich landscape for metaphorical analysis. While human moderators are associated with a wide range of metaphors – from “referees” to “protectors” and “helping hands” [24, 30] – AI moderation systems are often framed through adversarial and punitive metaphors such as “police officer” [25]. This highlights their typical use in identifying toxic players and punishing or removing them from the game.

Drawing from these intersecting bodies of literature, this paper leverages metaphors as creativity prompts to explore the opportunities and challenges involved in AI moderation design. Through a creative co-design approach, we aim to expand the conceptual vocabulary for AI moderation systems beyond current punitive frameworks while considering the ethical and practical implications of alternative approaches.

3 Method

We conducted four co-design workshops to explore participants' conceptualisations of AI moderation in multiplayer games. Workshop activities were centred around metaphor generation and application to facilitate design exploration. Each workshop lasted approximately 90 minutes and was conducted in-person at the Interaction Design Lab at The University of Melbourne following institutional ethics approval. Participants were compensated with an AU\$50 eVoucher each.

3.1 Participants and Recruitment

Participants were recruited through the researchers' academic networks (email and Discord), as well as the university's online noticeboard. Recruitment calls specified that the study sought (1) people who had played an online multiplayer game, and (2) professionals who had worked in the online multiplayer game space or on AI, including moderators, AI developers, and game developers. The final participant groups comprised of (1) One group of 5 HCI graduate researchers; (2) Two groups of 4 experienced multiplayer game players; and (3) One group of 4 AI policy, ethics, and technical experts.

3.2 Workshop Design

Following consent procedures, participants received an overview of AI moderation in gaming contexts (~10 minutes), including what it is, how and why it is used in gaming, and some contemporary examples. Participants then moved on to the first warm-up activity (~10 minutes), which asked them to individually brainstorm as many alternative metaphorical roles for AI in moderation as they could in 2 minutes, followed by a short sharing session.

In the second activity (~30 mins), in groups of two or three, participants were given a deck of 28 cards describing a range of alternative role metaphors categorised into three groups: caretaking & guidance, governing & overseeing, and support & management (see Appendix A.1). These metaphors and categories were created from the researchers' own brainstorming sessions and informed by previous work. They served to complement the organic, open-ended brainstorming in the first activity and help provide scaffolding for participants to draw on in their ideation. Participants were then invited to select one or two metaphors from this deck or from their own brainstorming to use as the foundation for designing their own alternative AI moderation systems. To ground their ideation, participants viewed a scenario video of a fictional battle royale game, 'Sorcerers Showdown' (~5 mins), created by the first author (see Figure 1). The game scenario featured four players on two competing teams using text chat, which in turn depicted cheating allegations, bullying, and spam. Researchers provided participants with paper, pens, markers, post-it notes, and guided design worksheet with optional prompts for participants to detail their design ideas (see Appendix A.2). This activity was followed by a whole group discussion.

Finally, the third activity (~30 mins) asked participants to reflect more deeply on the potential practical and ethical challenges that their ideas presented. Participants were provided with a red marker and asked to highlight problematic areas on their design worksheet. This activity was again followed by a whole group

discussion, and the workshop was concluded with a short discussion in which participants shared their parting thoughts (~5 mins). Collectively, these activities facilitated both individual ideation and group discussion, enabling participants to explore complex concepts through metaphors while maintaining focus on practical design applications.

3.3 Data Collection and Analysis

Data collection methods included audio and visual recordings of the workshops, as well as the gathering, transcription, and digitisation of all physical materials produced by participants, including their metaphor brainstorming lists, design worksheets and any other notes or illustrations made. The researchers also took observational notes during the workshops, and transcriptions were created of the discussion sessions of the workshops.

We adopted top-down thematic analysis to process the transcripts [5]. The first and second author collated and interpreted all collected materials to identify patterns in the participants' designs and discussions corresponding to the three pre-determined categories reflected in the metaphor card deck, producing three 'topic summary themes' in turn. All researchers collaboratively created initial codes using the coding software Dedoose, which were in turn honed through several meetings and discussions. This approach allowed us to synthesise insights from both recordings and written materials, providing a comprehensive understanding of participants' ideas and perspectives on AI moderation design.

4 Findings

During the initial brainstorming activity of our co-design workshops, the 16 workshop participants generated a total of 85 metaphors for novel AI moderation roles. We categorized these according to the three categories as devised for the cards: Caretaking & Guidance (28 instances – e.g. 'Therapist', 'Teacher'), Governing & Overseeing (25 instances – e.g. 'Spy', 'Big Brother'), and Assistance & Management (32 instances – e.g. 'Caretaker', 'Personal Assistant'). The Caretaking & Guidance category encompasses metaphors that frame AI moderation as a nurturing force, focusing on player development and community wellbeing. Governing & Overseeing metaphors position AI moderation as an authority figure, emphasizing top-down oversight. The Support & Management category includes metaphors that conceptualise AI moderation as a facilitative tool, helping communities and individual players manage their environments.

These metaphors were further classified by drawing from Desai and Twidale's [6] framework for metaphors types, with the majority being human-based metaphors (73 – e.g. 'Architect', 'Lifeguard'), followed by non-human (11 – e.g. 'Punching Bag', 'The Element of Water'), and fictional (2 – e.g. 'Ghost', 'Aquaman').

4.1 Spotlight Designs

Each group from the co-design workshops designed at least one AI moderation tool. Inspired by different metaphors, these designs showed a range of applications particularly along two dimensions: player interactivity and system impact. While some designs offered players little means to interact with the AI moderation system (low interactivity), others featured varied forms of interactivity in

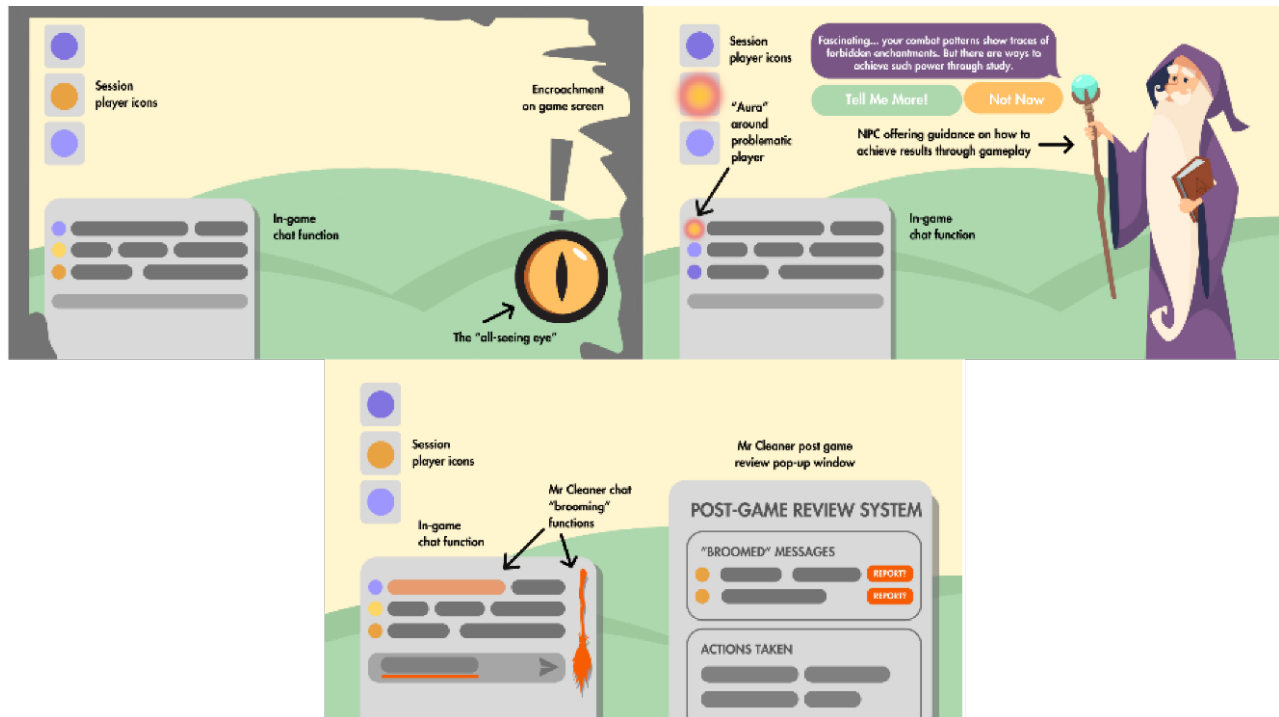


Figure 2: Interface mockups of the participant-led AI moderation systems derived from the co-design workshops. Each image depicts a player-facing screen view over the backdrop of a game. Top left: “A-Eye”; Top right: “The GAIMer”; Bottom: “Mr Cleaner”.

which the players could control or act autonomously to customise moderation processes (high interactivity). Similarly, while some moderation system designs had minimal impact on the game itself (low impact), others had the AI interacting directly with elements of the user or gaming interface in response to problematic behaviour (high impact). In this section, we profile three designs, chosen to illustrate a range of possibilities across our analytical dimensions while highlighting innovative approaches to common moderation challenges (see Figure 2).

4.1.1 Design 1: A-Eye. Drawing from the ‘All-seeing eye’ metaphor, the *A-Eye* concept envisions an all-seeing AI system for monitoring and moderating online game behaviour. This low-interactivity/moderate-impact system, represented as a visible open eye on the gaming interface, would access all in-game actions and data and work to identify problematic behaviours. Rather than employing traditional punitive measures, *A-Eye* operates through subtle environmental adaptations. The interface would automatically adapt to player interactions in real-time depending on their severity and frequency, potentially using subtle cues such as voice or soundscape elements to influence conduct or quietly removing chat features. As one participant noted, “people behave differently when they feel like they’re being watched,” highlighting the system’s potential functioning as a kind of ‘panopticon’. The design team envisioned *A-Eye* as a preventative rather than reactive system, emphasizing de-escalation and behaviour modification through environmental cues.

However, the *A-Eye* concept raised concerns about player privacy, trust, and misuse. Participants acknowledged that players might feel frustrated or violated by the system’s interventions on screen. As one participant suggested, players may also “do bad things . . . just to see this change in the interactiveness of the screen.” Participants noted potential technical limitations regarding the system’s ability to process and respond to millions of simultaneous player interactions in real time.

4.1.2 Design 2: The GAIMer. The *GAIMER* system focuses on addressing cheating through in-game consequences that align with a game’s narrative and/or mechanics. Drawing from the ‘Non-player character (NPC)’ metaphor, this moderate-interactivity/high-impact system acts as an in-game deity that discourages cheating while maintaining the game’s context. The system proposes various in-game penalties for cheaters to “make obsolete whatever benefit” cheats bring, such as increased attack/spell backfire risks, debuffs, or indicators attached to their avatar, with one designer suggesting cheaters could have “an aura around them, without being too publicly shamed”. This approach aimed to keep players engaged even after punishment, unlike current moderation methods that can result in “rage-quitting” or system-enforced bans. The *GAIMER* system also incorporated a teaching element, where instead of merely punishing cheaters, it offered guidance to affected players on how to achieve similar results through legitimate gameplay.

The participants expressed some concerns about potentially patronising players or inadvertently encouraging cheating by making

it more appealing to players who wish to challenge themselves or explore the game by taking on moderation penalties. They also acknowledged social challenges, such as the potential for “witch hunts” of cheating players, and identified technical challenges given the ongoing “arms race” between cheating software and anti-cheat measures.

4.1.3 Design 3: Mr Cleaner. *Mr. Cleaner* is a high-interactivity/high-impact moderation software based on the “Cleaner” metaphor that is designed for chat functions in online multiplayer games, with both autonomous and player-controlled features. It can sanitise in-game chat by “delet[ing] spam or negative words or toxic players” while also allowing players to “broom” or “sweep” away unwanted content in real time. The system includes features to prevent players sending toxic messages and to detect spam.

Participants identified several challenges, including the time-consuming nature of constant “brooming” and the risk of accidentally removing important messages. To address concerns about accuracy and transparency, they proposed a post-game review box showing ‘cleaned’ content and allowing further action. While this system offers flexibility across various games with chat functions, the designers noted it wouldn’t work well with voice chat. They also recognised potential privacy concerns and the risk of overuse or disruptive reporting, which could increase human moderators’ workload.

4.2 Alternative AI Moderation: Functions, Perceptions and Challenges

In this section we further respond to RQ1 – how stakeholders conceptualise alternative AI moderation through metaphors – with the results of our thematic analysis. Below, we present three key themes and detail the functions, perceptions and challenges associated with different approaches to moderation design.

4.2.1 AI as Censor and Responder: Monitoring, Punishment and Enforcement. Many designs reframed or remixed the monitoring and enforcement capacities of existing AI moderation systems. For example, censoring was a key function in some of the interfaces where the AI would prevent certain phrases or words from being sent or clear messages automatically (as in *Mr Cleaner*). Despite encouragement to envision alternative systems, punishment remained a key function in many of the designs that players saw as valuable and even necessary. At the same time, participants considered a range of novel punishments systems. Some innovative approaches integrated such responses directly into the game mechanics. *The GAIMER* design, for example, gave cheating players in-game disadvantages or “debuffs”, such as weakening their player character’s spell strength, damaging their health, or increasing the likelihood of spells backfiring. Another example proposed that interactions with in-game non-playable characters would be influenced by problematic in-game behaviour, much like in role-playing games (RPGs) such as *Baldur’s Gate 3*, where in-game vendors are aware of players’ actions in the world and respond accordingly. Others suggested that players could have in-game resources taken away, but ethical issues arose regarding the potentially disproportionate impact on players depending on “how rich players are” in the game. Interestingly,

only one design suggested a reward system for good behaviour – even when reimagining moderation, participants gravitated toward corrective rather than inventive-based approaches.

4.2.2 AI as Coach and Counsellor: Education, Support and Rehabilitation. Some designs envisioned AI moderators taking on nurturing and educational roles, moving beyond pure enforcement. These approaches generally fell into three categories: coaching for prevention, intervention for correction, and support for victims. Participants proposed AI systems that could proactively guide or coach players in a personalised fashion to “engage healthily” with the player community. For example, AI-powered NPCs could ask players to reflect on their problematic behaviour or offer support if they noticed players were “withdrawn” or angry, helping players develop emotional awareness and self-regulation skills. Participants also showed several innovative approaches to behavioural correction. For example, some designs suggested the AI tool may sign players up to a “rehabilitation or remedial class”, make them do some form of in-game community service, or educate cheating players on how to achieve high-level performance without cheating via prompts. One group even suggested an NPC might “start to cry” to make harms more concrete and “induce feelings of guilt” in the toxic player. Drawing from the ‘protector’ metaphor, some participants designed systems specifically focused on supporting harassment targets, reflecting a significant departure from moderation systems that primarily focus on punishing offenders. One group suggested that the AI could send them “tools and strategies to recover,” such as references to relevant helplines. However, participants identified some critical challenges with supportive approaches, including potential resistance from players and AI’s (un)suitability for giving emotional support.

4.2.3 AI as Assistant and Facilitator: Reporting, Assistance and Resolution. Participants were also drawn to assistive metaphors, understanding AI as useful in making the process of reporting and moderation easier for human players and moderators. For instance, participants suggested designs in which the AI could collect evidence of harms for players to submit in reports, draft email templates to customer service, or feed into an interface that allows players to keep track of reports and make case management easier. In this way, AI was seen as well positioned to “take away some of the burden” of players having to make and manage (and receive) numerous reports in what can be fast-paced gameplay. Participants also suggested that the AI could function as an assistant to human moderators, offering them response and remediation options while assisting in prioritisation of cases, such as through organising reports according to severity. Reflecting on concerns that AI would be a poor replacement for human decision-making and engagement in moderation, participants espousing these kinds of designs saw humans as necessary in making final moderation decisions on whether behaviour was “toxic or not”.

5 Discussion

Our workshop participants demonstrated a wide array of creative alternative approaches to AI moderation design, drawing from existing mechanisms based on detection and enforcement while introducing new and underexplored approaches based on education,

support and assistance. Although they were encouraged to imagine functions beyond more punitive roles, many designs and discussions still drew from ‘governing and overseeing’ metaphors. This reflects previous findings that suggest players often have a dominant mindset associated with punitive justice models in games even though such models fail to address their needs [31]. Nevertheless, the variations participants introduced to these approaches highlighted unexplored potential regarding AI moderation, particularly along the axes of interactivity and impact. This included interface integration (*A-Eye*), game and narrative integration (*The GAIMER*), transparency and autonomy (*Mr Cleaner*), and more. Participants also raised ethical issues with these approaches, highlighting existing concerns around AI moderation’s potential for invasiveness, opacity, manipulation [10] and exploitability. These persistent concerns illustrate the importance of centring user trust and privacy in any moderation system, and that alternative designs have the potential to both ameliorate and exacerbate these concerns. We reflect on our participants’ designs and considerations in the next section, highlighting three avenues for future AI moderation design and research.

5.1 Future Design and Research Directions

5.1.1 From Punitive to Supportive Frameworks. Our analysis revealed an interest in caretaking and guidance metaphors. These were both individualized (e.g. “teacher”, “therapist”, “personal trainer”) as well as community-focused (e.g. a “caretaker” who is “beneficial to the community”). We emphasise the potential for leveraging AI moderation toward player/community development and growth, as found in support tools and restorative justice approaches [20, 31]. Broader approaches based on support have potential to not only decrease stress and increase feelings of control [20] but also foster less toxic player communities [29]. Challenges that must be considered in these approaches include privileging some users over others [22], player skepticism [20], and the suitability of AI in performing tasks associated with expert health professionals [7].

5.1.2 Player Agency and Control. A critical limitation of traditional punitive approaches to player moderation is their positioning of players as passive subjects under surveillance and monitoring. Reflecting previous work that found “players desired to be involved in the governance process” [17], many alternative metaphors emerging from our research reframe players as active participants in the online gaming ecosystem, as exemplified in the *Mr Cleaner* design. Meanwhile, more controlling metaphors like ‘puppeteer’ (noted as “bad idea, control maybe too much”) or ‘spy’ were associated with negative implications, highlighting a broader unease with AI having too much power in online governance and beyond [8]. Finding an appropriate balance between human moderators and AI intervention is a key goal for online governance [2, 23], and there is potential to explore how players themselves can participate in and contribute to moderation decisions.

5.1.3 Contextual Integration. Designs such as *The GAIMer* pointed to possibilities for weaving moderation systems naturally into game mechanics and narrative structures. Echoing calls for the integration of moderation tools into the game client [20], there is potential

for moderation systems to be designed to feel native to their respective game environments rather than externally imposed. Such integration could reduce the perceived friction between ‘in-game’ play, often associated with a “competitive ethos”, and ‘out-of-game’ moderation demands [16, 26]. Further research can explore what forms of integration are successful, recognising the potential for inadvertently trivialising moderation or reinforcing problematic behaviour.

5.2 Limitations and Conclusion

This study reflects an initial exploration into player and expert conceptualisations of AI moderation in multiplayer games. Given the small number of workshops and participants, as well as the workshop scenario’s focus on a team-based competitive game, the findings have limitations in scope and application. The feasibility of these designs, as well as their impact on user trust and privacy, require more detailed attention. Future work may examine the perception and impact of alternative moderation styles on player behaviour through broader scale, mixed methods and experimental approaches. We hope that this work inspires further explorations into the design of AI moderation for online governance.

Acknowledgments

We are grateful to the Centre for AI and Digital Ethics (CAIDE) at The University of Melbourne for providing seed funding for this project. We would also like to thank our generous participants for their time and insights.

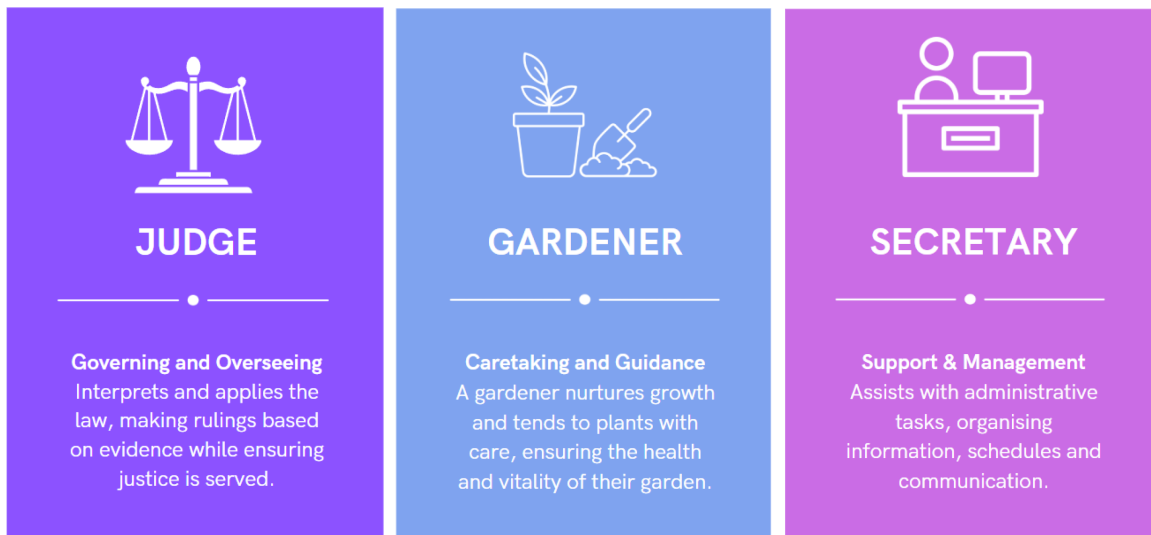
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A APPENDICES

A.1 Examples of metaphor role cards provided to participants in the second activity of the co-design workshops.



A.2 Design worksheet with design prompts for the second activity of the co-design workshops.

The worksheet is titled "GAIM CO-DESIGN WORKSHOP" and includes a logo. It is divided into several sections for design prompts:

- Metadata:** DATE: _____, NAME/S: _____, CARD: _____, NAME OF AI TOOL: _____.
- USER STORY:** A 2x4 grid of boxes with arrows indicating a flow between them.
- WHAT DOES IT DO? / HOW DOES IT DO IT? / WHO CAN INTERACT WITH IT?:** Three vertical columns for defining the tool's function, mechanism, and users.
- INTERFACE DESIGN:** Two large rounded rectangular boxes for sketching the interface.
- HOW IT IS EXPERIENCED BY:** Four circles labeled "PLAYERS...", "MODERATORS...", "DEVELOPERS...", and "OTHER..." for mapping user experiences.